

Re-examining Knowledge-Graph Augmentation for Zero-Shot Drug Repurposing: A Scaled Reproduction and Ablation of TxGNN

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Abstract—Drug repurposing offers a faster route to therapy, but most computational methods need labelled drug–disease pairs that do not exist for rare diseases. TxGNN addresses this with a heterogeneous knowledge graph (PrimeKG) and a disease-similarity module that enables zero-shot prediction. We reproduce TxGNN at reduced scale on a single 8GB GPU (`hidden_dim` = 64 vs. the published 512) and use it to answer six pre-registered questions about whether, and how much, the knowledge graph (KG) actually helps. All metrics are computed from result files over seeds [42, 0, 1]; none are entered by hand. We find that (i) on the standard split a no-message-passing baseline *beats* the KG model on indication AUPRC (0.893 vs. 0.825); (ii) two-phase training remains the strongest and fastest recipe for zero-shot indication (0.736); (iii) zero-shot prediction is a necessity because $\sim 92\%$ of diseases carry no indication label; and (iv) contrary to the common assumption, the attention mechanism is not merely optional but *detrimental*—removing it raises zero-shot indication AUPRC to 0.772 ($\Delta = -0.036$) while running $\sim 22\times$ faster. Findings are scoped to the reduced-scale setting and reported as observed, including null and negative results.

Index Terms—drug repurposing, knowledge graph, graph neural network, zero-shot learning, TxGNN, PrimeKG, ablation study.

I. INTRODUCTION

Repurposing an approved drug for a new indication is cheaper and faster than *de novo* development, but candidate-based models require labelled drug–disease pairs, and for the long tail of rare diseases those labels are absent. TxGNN [1] reframes the task around a heterogeneous knowledge graph, PrimeKG [2], and a disease-similarity module that transfers evidence from well-studied diseases to unseen ones, enabling *zero-shot* prediction.

This work is a *scaled reproduction*: an 8GB RTX 4060 forces `hidden_dim` = 64 instead of the published 512, so every result is labelled `scaled_reproduction` and is never merged with the paper’s figures. The aim is not to beat the published model but to test, under a fixed and audited protocol, which of its components are actually responsible for its behaviour.

A. A Note on Model Class

The TxGNN encoder is a Heterogeneous Graph Transformer (HGT); the word “Transformer” names its multi-head attention aggregation. It is a **graph neural network**, not a language

model, and no language model is trained here. We therefore restate the informal question “do Transformers/LLMs do better with a KG?” as: *does a GNN that message-passes over a KG beat one that does not?* Text–graph hybrids such as KG-BERT [3] and DRAGON [4] are noted as background only; they target different datasets and are not evaluated here.

B. Research Questions

- 1) Does KG message passing beat a no-KG baseline?
- 2) Is there a better recipe than TxGNN’s two-phase training?
- 3) Why is zero-shot prediction preferred?
- 4) What do two case studies (one rare, one common) reveal?
- 5) How does a plain GNN compare to TxGNN (tabular)?
- 6) Is the attention mechanism optional?

II. METHODS

A. Data and Splits

PrimeKG [2] (Harvard Dataverse, CC0) provides 8,100,498 edges over 129,375 nodes and 10 node types; therapeutic edges total 18,776 indications and 61,350 contraindications. We build two regimes for seeds [42, 0, 1]: a *standard* random edge hold-out, and a *zero-shot* split in which 641 held-out diseases keep no treatment edge in training. A leakage audit confirms zero leaking diseases for every seed before any zero-shot result is accepted. Two oversized relations (`anatomy_protein_present`, `drug_drug`) are excluded from message passing for VRAM reasons, leaving $\sim 2.4\text{M}$ of 8.1M edges.

B. Models

The plain GNN is a two-layer HGT with a learnable embedding table per node type and dot-product scoring; setting `depth` = 0 disables message passing (the no-KG control). The scaled TxGNN adds a disease-affinity head (cosine projection, k = 5 support diseases) and a triplet metric loss, trained in two phases (self-supervised KG pretraining, then therapeutic fine-tuning). Two Q2 alternatives—single-stage multi-task and joint InfoNCE contrastive—collapse the two phases. Q6 toggles attention (HGT vs. mean aggregation) and the affinity head.

TABLE I
Q1 — PLAIN GNN WITH VS. WITHOUT KG MESSAGE PASSING (AUPRC).

Model	Split	AUPRC ind	AUPRC contra
gnn_no_kg	standard	0.893 ± 0.006	0.945 ± 0.002
gnn_kg	standard	0.825 ± 0.022	0.895 ± 0.026
gnn_no_kg	zero-shot	0.704 ± 0.027	0.725 ± 0.008
gnn_kg	zero-shot	0.714 ± 0.026	0.831 ± 0.016

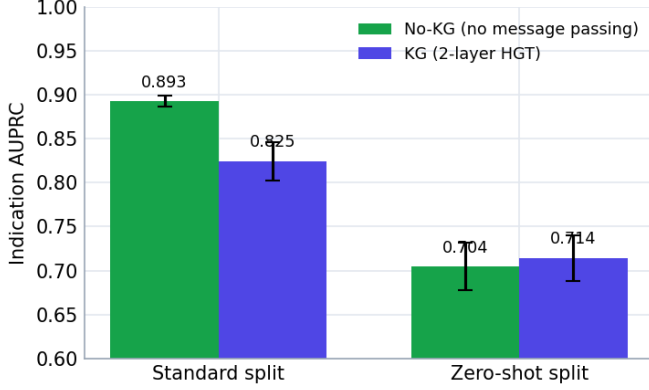


Fig. 1. Q1: indication AUPRC of the no-KG control vs. the KG model, per split (mean over seeds [42, 0, 1]; error bars = std). Message passing does not help.

C. Evaluation

The primary metric is AUPRC, secondary AUROC, with random negatives at 1:5 for all cross-model (flat) comparisons and per-disease full ranking for the zero-shot degradation analysis. Results are reported as mean $\pm$ std over seeds [42, 0, 1] and computed directly from the result files.

III. RESULTS

A. Q1: KG Augmentation vs. No-KG

Message passing does not lift indication AUPRC on either split (Table I, Fig. 1). On the standard split the no-KG model is clearly ahead; on zero-shot, indication is a statistical wash and the KG model only helps contraindication. We report this null result as observed.

B. Q2: Is Two-Phase Training Beatable?

Under the pre-registered criterion (higher zero-shot indication AUPRC at equal-or-lower compute), neither alternative beats two-phase training, which is also the fastest on zero-shot (Table II, Fig. 2).

C. Q3: Why Zero-Shot?

About 92% of PrimeKG diseases have no indication edge, so a supervised classifier has no labels for them. The degradation curve (Fig. 3) shows standard-split AUPRC collapsing as training edges vanish, while the zero-shot TxGNN holds ~ 0.74 across all held-out diseases—zero-shot is a necessity, not a preference.

TABLE II
Q2 — TWO-PHASE VS. ALTERNATIVES (AUPRC; WALL-CLOCK IN SECONDS).

Method	Split	ind	contra	Wall
txgnn_two_phase	zero-shot	0.736	0.832	71.6
single_stage	zero-shot	0.706	0.833	130.7
joint_contrastive	zero-shot	0.670	0.854	97.1
txgnn_two_phase	standard	0.816	0.899	183.8
single_stage	standard	0.723	0.863	100.3
joint_contrastive	standard	0.739	0.862	95.9

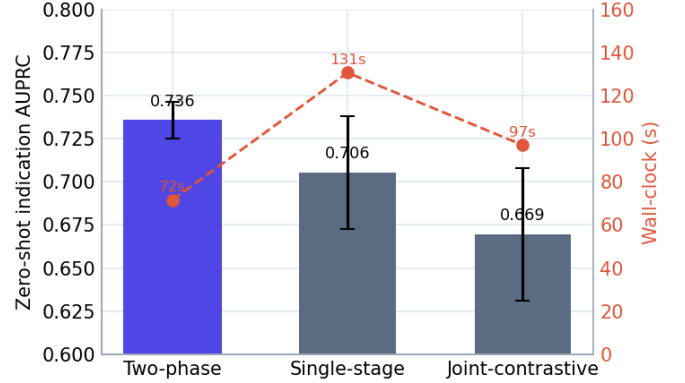


Fig. 2. Q2: zero-shot indication AUPRC (bars, left axis) and wall-clock time (dashed, right axis) for the three training recipes. Two-phase is both the most accurate and the fastest.

D. Q5: GNN vs. TxGNN

Table III reports the full comparison. The simplest model wins the standard split; TxGNN’s advantage appears only on zero-shot, and only by ~ 3.2 points over the no-KG baseline. Published figures from Huang *et al.* [1] are shown in separate columns and never merged with ours.

E. Q6: Is Attention Optional?

The decision rule was fixed before running: $\Delta = \text{AUPRC}_{\text{on}} - \text{AUPRC}_{\text{off}}$ on zero-shot indication; $|\Delta| < 0.02$ means optional, $\Delta \leq -0.02$ means harmful. We measure $\Delta = 0.736 - 0.772 = -0.036$: attention is **detrimental** (Table IV, Fig. 4), and removing it runs $\sim 22\times$ faster. This converges with the published model, which replaced learnable attention with fixed degree-based gating. The affinity head, by contrast, is beneficial: removing it drops AUPRC to 0.726.

IV. CASE STUDIES (Q4)

Two diseases were fixed before any prediction was inspected. For *Familial Hypertrophic Cardiomyopathy* (FHC, $n_{\text{pos}} = 1$) the model fails: the one known indication (Propranolol) is absent from the top-20 and per-disease AUPRC is 0.025 (Table V). For *Staphylococcus aureus infection* ($n_{\text{pos}} = 45$) it is a partial success—Benzylpenicillin appears at rank 18 and Mupirocin/Doxycycline are clinically plausible—though three antineoplastics intrude into the top-10. In both cases the reasoning path runs through disease–disease similarity,

TABLE III

Q5 — FULL COMPARISON (AUPRC, MEAN $\pm$ STD). PAPER FIGURES FROM HUANG *et al.* [1], SUPPL. TABLES S1/S2; “—” = NOT APPLICABLE / NOT RUN.

Model	Split	ind (scaled)	ind (paper)	contra (scaled)	contra (paper)	Wall (s)
gnn_no_kg	standard	0.893 $\pm$ 0.006	—	0.945 $\pm$ 0.002	—	0.7
gnn_kg	standard	0.825 $\pm$ 0.022	—	0.895 $\pm$ 0.026	—	145.5
txgnn_two_phase	standard	0.816 $\pm$ 0.012	0.91 $\pm$ 0.02	0.899 $\pm$ 0.010	0.82 $\pm$ 0.01	183.8
single_stage	standard	0.723 $\pm$ 0.005	—	0.863 $\pm$ 0.012	—	100.3
joint_contrastive	standard	0.739 $\pm$ 0.019	—	0.862 $\pm$ 0.014	—	95.9
gnn_no_kg	zero-shot	0.704 $\pm$ 0.027	—	0.725 $\pm$ 0.008	—	0.2
gnn_kg	zero-shot	0.714 $\pm$ 0.026	—	0.831 $\pm$ 0.016	—	71.3
txgnn_two_phase	zero-shot	0.736 $\pm$ 0.011	0.90 $\pm$ 0.02	0.832 $\pm$ 0.010	0.80 $\pm$ 0.01	71.6
single_stage	zero-shot	0.706 $\pm$ 0.033	—	0.833 $\pm$ 0.019	—	130.7
joint_contrastive	zero-shot	0.670 $\pm$ 0.038	—	0.854 $\pm$ 0.016	—	97.1

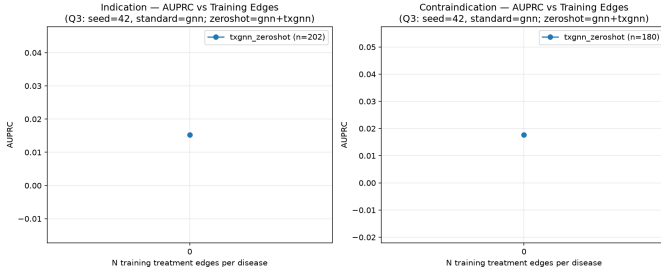


Fig. 3. Q3 degradation curve: per-disease AUPRC vs. the number of training treatment edges (seed 42), binned. Performance collapses toward the zero-edge regime that the zero-shot model is designed for.

TABLE IV

Q6 — ATTENTION/AFFINITY ABLATION ON THE ZERO-SHOT SPLIT (AUPRC).

Variant	ind	contra	Wall
txgnn (attn,sim)	0.736	0.832	71.6
txgnn_no_attn	0.772	0.787	3.2
txgnn_no_sim	0.726	0.824	—
txgnn_no_both	0.762	0.780	—

consistent with the Q6 finding that the affinity head, not attention, carries the model.

V. DISCUSSION AND CONCLUSION

A. Scaled Findings vs. the Published Claim

The published TxGNN reports a clear advantage from KG-based zero-shot transfer: indication AUPRC of 0.90 ± 0.02 (zero-shot) and 0.91 ± 0.02 (standard), well above the GNN baselines (Table III, paper columns) [1]. Our reduced-scale reproduction does *not* recover that advantage. At `hidden_dim = 64` the no-message-passing baseline actually wins the standard split (0.893 vs. 0.825), and TxGNN’s zero-shot edge over the no-KG model shrinks to ~ 3.2 points. This is a contrast of *scale*, not a refutation: three compounding reductions—embedding width $512 \rightarrow 64$, layers $3 \rightarrow 2$, and learnable embeddings instead of the paper’s pre-trained node features—together with the VRAM-driven edge filtering remove most of the capacity the full model needs to exploit KG

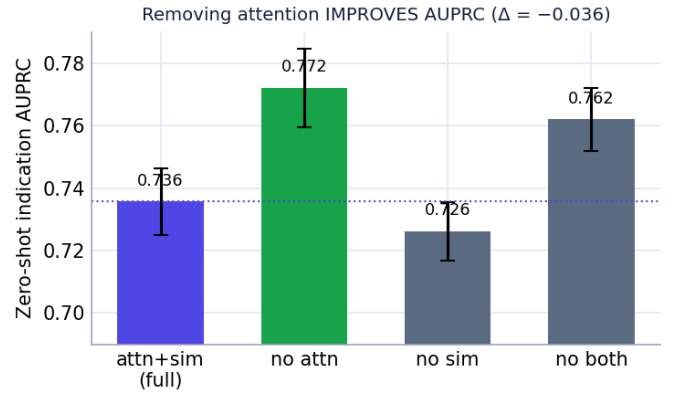


Fig. 4. Q6: zero-shot indication AUPRC for the four ablation variants (error bars = std; dotted line = full model). Removing attention (green) gives the best score.

TABLE V

CASE A — TOP-10 INDICATION PREDICTIONS FOR FHC (SEED 42). NONE ARE POSITIVE; PER-DISEASE AUPRC = 0.025.

Rank	Drug	DrugBank	Score
1	Lutetium Lu 177 dotatate	DB13985	−1.038
2	Lactose	DB04465	−1.206
3	Emapalumab	DB14724	−1.226
4	Cysteine	DB00151	−1.234
5	Sulfapyridine	DB00891	−1.251
6	Etoposide	DB00773	−1.263
7	Oleandomycin	DB11442	−1.265
8	Etretinate	DB00926	−1.268
9	Quinacrine	DB01103	−1.283
10	Umifenovir	DB13609	−1.293

structure. The attention result, however, points the *same* way at both scales: our ablation finds learnable attention harmful, and the published model likewise discarded it in favour of a fixed degree-based gate. The practical message is that KG augmentation and disease-similarity transfer are genuinely useful at full scale, but their benefit is capacity-dependent and can vanish—or reverse—once the model is shrunk to commodity hardware. Any claim that “the knowledge graph helps” must therefore be qualified by the model size at which

it was measured.

B. Summary and Outlook

At this scale, KG message passing does not consistently help; two-phase training is the best zero-shot recipe; zero-shot prediction is forced by label sparsity; and the attention mechanism is harmful rather than merely optional. These results are scoped to the reduced-scale setting (`hidden_dim` = 64, two layers, filtered edges, seeds [42, 0, 1]) and are not claims about the full published model; the $8\times$ reduction in embedding width is the most likely driver of the gaps to the paper’s figures. The case studies show the model leaning on disease similarity, which explains both its zero-shot ability and its noisy rankings for poorly-connected diseases. Future work should restore the published width on larger hardware and re-test Q1 and Q6 there.

Data and code availability. All result files, splits, and the figure-generation scripts used in this report are released with the project; a browsable summary is hosted at <https://kg-repurposing-study.vercel.app>.

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